James Litchfield

Product Engineer | TypeScript, React, Node.js Ruby on Rails
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Summary

Product engineer with 6+ years of experience building web products in TypeScript, React, Node.js and Ruby on Rails. I own features from the database to the browser, with a focus on accessibility, performance and reliability. My favourite work removes a bottleneck for someone else, like a CMS marketing can publish from without engineering help. I share what I learn through code review, pairing and technical showcases, and use AI tooling to speed up parts of my workflow.

Skills

Languages: TypeScript, JavaScript (ES6+), Ruby, HTML, CSS

Frontend: React (function components, hooks), Next.js (ISR), Vite, Tailwind CSS

Backend: Node.js, Ruby on Rails, Supabase, PostgreSQL (Postgres), NestJS, Sidekiq, GraphQL, REST APIs, MongoDB

Testing: Cypress, Playwright, Storybook, Chromatic, RSpec, automated testing, A/B testing

Accessibility and performance: Accessibility (WCAG 2.2), Core Web Vitals, performance optimisation, Lighthouse, PageSpeed Insights, Cloudinary, search engine optimisation (SEO)

Delivery and monitoring: Git, CI/CD (GitHub Actions), Bugsnag, Heap

AI tooling: Cursor, Claude, GitHub Copilot, MCP servers

Earlier experience: Python, Swift, SwiftUI, PHP, WordPress, C#, Xamarin

Work Experience

Career break | May 2026 - Present

Made redundant from Scan.com in May 2026. Took time out to travel and am now looking for my next role.

Product Engineer | Scan.com | London (remote) | November 2022 - May 2026

Medical imaging scale-up operating across the UK and US, with 200+ UK scanning centres.

- Designed and built a CMS (Rails, React) that lets marketing publish SEO landing pages and trial new products without engineering support, saving at least 1 engineer-day per page.
- Built internal ops tooling, payment integrations (Stripe) and validation workflows for the booking platform (Rails, React, TypeScript, Postgres, Tailwind), and tested cross-site data changes to prevent regressions.
- Codified the team's established practices into an AI-assisted review workflow (Claude and Notion MCP), so a review agent could check changes against them.
- Designed and shipped an embeddable booking widget (embed script plus iframe) serving a slimmed-down booking flow on partner domains, using cross-window messaging to persist booking state across refreshes and redirects. Wrote the integration docs for the first partner to onboard.
- Owned the redesign of fullbodyscan.com (Rails) in a two-person pod. Set the page architecture and UI, and rebuilt the site to improve accessibility against WCAG 2.2. Conversion rate rose and bounce rate fell in a before-and-after comparison (measured in Heap).
- Ran production support and on-call triage across the Rails and React applications. Used Bugsnag and Sidekiq to debug errors and failed jobs, and followed issues through to resolution.
- Performed code reviews on pull requests from all engineers on the team and paired regularly with other developers to share knowledge and spread team practices.
- Improved Core Web Vitals across Scan.com's connected domains, including a migration to Cloudinary for image optimisation. Real-user LCP (measured in PageSpeed Insights) dropped from ~4s to under 2s, and CLS fell to 0.01.

- Introduced modern browser APIs and progressive enhancement patterns, cutting client-side JavaScript. Worked with design to document these patterns in a component library (Vite, Storybook) and presented it to the engineering team in technical showcases.

Mid-Level Web Developer | Dexerto | London (remote) | January 2021 - November 2022

Scale-up news outlet covering gaming and internet culture, with 30M monthly visitors in 2022.

- Redesigned dexerto.com (Next.js, TypeScript, GraphQL, Tailwind). Improved SEO with incremental static regeneration, structured data and sitemap changes, which lifted organic traffic.
- Implemented infinite scroll across article pages, contributing to a ~20% increase in pages per session and ~10% increase in Google ad impressions, confirmed in an A/B test.
- Wrote custom PHP plugins that extended the headless WordPress GraphQL API, exposing custom post fields and ad slots to the Next.js frontend.
- Worked with design to build the React component library behind the Next.js UI of mystats.com (tables, charts, live stats, maps).
- Set up automated testing across all domains with Storybook, Chromatic, Playwright and Cypress. The tests covered core components and critical user flows, and caught visual regressions before release.

Full-Stack Developer | Eledev Digital | Remote | May 2020 - January 2021

Early-stage start-up specialising in mobile app development.

- Started as a freelancer in May 2020 and moved to a full-time role in July 2020.
- Co-developed the Hour Blocks (Swift, SwiftUI) iOS app, which received 100,000+ downloads and Apple "App of the Day" in 150+ countries.
- Built the NestJS and MongoDB backend for Pro Wrestling Simulator online persistence, supporting 300+ registered users.

Junior Developer | Kmatic | Liverpool | May 2020 - July 2020

Company providing customisable solutions for the collection, analysis and processing of geospatial data.

- Delivered features for a Xamarin mobile app and internal data-processing tools, and assessed ARKit/ARCore feasibility.

Education

BSc Computer Science (1st class) | University of Liverpool | 2017 - 2020

Projects

Reelette | October 2026 - Present

A collaborative movie-night app using The Movie Database, featuring shared watchlists, real-time updates and filtered random selection, with database-level access controls.

GitHub: https://github.com/james-litch/reelette

URL: https://reelette-red.vercel.app/

Tools: Next.js, TypeScript, Supabase, Tailwind CSS, OpenAI Codex

Secret Santa | October 2026 - Present

Built a private gift-exchange app, with fair random assignments, secure participant access and retry-safe operations. Tested concurrency, accessibility and connection failures through integration and browser tests.

GitHub: https://github.com/james-litch/SecretSanta

URL: https://secret-santa-pink-psi.vercel.app/

Tools: Next.js, TypeScript, Redis, Tailwind CSS, OpenAI Codex